

Data Preprocessing

Dataset: Titanic dataset

Tools / Libraries: Python , Pandas, NumPy, Scikit-learn (sklearn.preprocessing, sklearn.impute), Matplotlib/Seaborn (for visualization)

1. Data Cleaning

- Load the dataset and inspect shape, datatypes, and summary statistics
- Identify and fix inconsistent entries
- Drop irrelevant columns that add no predictive value (e.g., PassengerId, Ticket)

2. Handling Missing Values (Imputation)

- Identify columns with missing values (Age, Cabin, Embarked) and count them
- Handle missing values using:
 - dropna() for columns with excessive missingness (e.g., Cabin)
 - fillna() or SimpleImputer with mean/median for Age
 - fillna() or SimpleImputer with mode for Embarked

3. Duplicate Removal

- Check for duplicate rows using duplicated()
- Remove duplicates using drop_duplicates() and report row count before and after

4. Outlier Detection

- Detect outliers in Fare using the IQR method (or Z-score method)
- Visualize outliers using a boxplot before and after treatment
- Remove or cap detected outliers, with justification

5. Encoding Categorical Variables (Types)

- Apply **Label Encoding** on Column 'Sex'
- Apply **One-Hot Encoding** on Column 'Embarked'

6. SAVE FINAL CLEANED DATASET